LaTeX Template

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Is it that you're being good to-get-her, or that you think you're good together?

Me (inspired by "Good Together" — Shy Martin)

Contents

§1 `tcolorbox` Environments

This is the color of the default **bold text**.

Definition 1.1 (Name)

This is a Definition.

Theorem 1.2 (Name)

This is a Theorem.

Lemma 1.3 (Name)

This is a Lemma.

Corollary 1.4 (Name)

This is a Corollary.

Proposition 1.5 (Name)

This is a Proposition.

Notation:

- Notation 1.
- Notation 2.
- Notation 3.

Question 1.6 (Name). This is a Question.

Exercise 1.7 (Name). This is an Exercise.

Hint: 1

Problem 1.8 (Name). This is a Problem.

Hints: 3 2

Algorithm 1.9 (Name)

This is an Algorithm.

Claim 1.10 (Name) — This is a Claim.

Proof. This is a Proof. □

Example 1.11 (Name)

This is an Example.

Solution. This is a Solution. ■

Remark 1.12 (Name)**.** This is a Remark.[a]

[a] This is footnote for this remark.

§2 Using references

Here is how you can use the references made above: Definition 1.1, Theorem 1.2, Lemma 1.3, Corollary 1.4, Proposition 1.5, Question 1.6, Exercise 1.7, Problem 1.8, Algorithm 1.9, Claim 1.10, Example 1.11, Remark 1.12.

Please ignore the linter errors here (if you're using a linter). The linter does not recognize references like `theorem:firstthm` since there is no explicit reference as such. However, the code compiles completely fine since `tcolorbox` sets up references by adding the environment name before the reference label.

§3 Other Environments

Col1	Col2	Col3
1	2	3

Table 1: This is a table

(a) First.

(b) Second.

- First.
- Second.

Figure 1: Cool Gojo 4K Wallpaper

§4 Math Environments

$$\begin{aligned} x^2 &= \frac{2x^2}{2} \\ &= \frac{(x+1)^2 + (x-1)^2 - 2}{2}. \end{aligned}$$

$$f(x) = \begin{cases} 0, & \text{if } x \text{ is rational} \\ 1, & \text{otherwise.} \end{cases}$$

$$g(x) = \begin{cases} 0, & \text{if } x \text{ is irrational} \\ \frac{1}{q}, & \text{if } x = \frac{p}{q} \text{ where } p,\ q \text{ are integers with } \gcd(p, q) = 1. \end{cases}.$$

$$\sin \cos \tan \operatorname{cosec} \sec \cot \arcsin \arccos \arctan \operatorname{arccsc} \operatorname{arcsec} \operatorname{arccot}.$$

$$\nu_p(p^2) = 2 \text{ and } \operatorname{Pow}\odot(ABC)(A) = 0 \text{ and } \operatorname{ord}_p((p-1)!) = 2.$$

$$\iint\limits_{x^2+y^2\le 1} (x+y)\mathrm{d}x\mathrm{d}y = 0.$$

$$\int_0^1 \ln(x) = -1.$$

$$\frac{\mathrm{d}^n f}{\mathrm{d}x^n} \frac{\partial^n f}{\partial x^n}.$$

§5 CS and ASY Environments

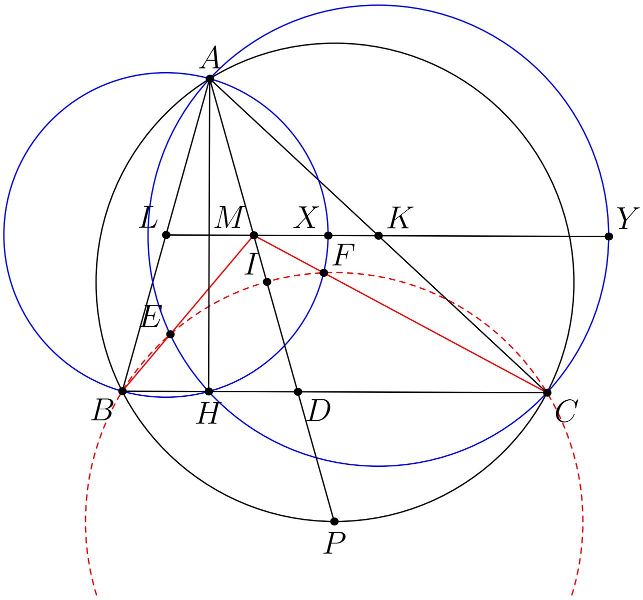

Figure 2: An asy diagram

```
import numpy as np

def incmatrix(genl1,genl2):
    m = len(genl1)
    n = len(genl2)
    M = None #to become the incidence matrix
    VT = np.zeros((n*m,1), int)  #dummy variable

    #compute the bitwise xor matrix
    M1 = bitxormatrix(genl1)
    M2 = np.triu(bitxormatrix(genl2),1)

    for i in range(m-1):
        for j in range(i+1, m):
            [r,c] = np.where(M2 == M1[i,j])
            for k in range(len(r)):
                VT[(i)*n + r[k]] = 1;
                VT[(i)*n + c[k]] = 1;
                VT[(j)*n + r[k]] = 1;
                VT[(j)*n + c[k]] = 1;

                if M is None:
                    M = np.copy(VT)
                else:
                    M = np.concatenate((M, VT), 1)
```

```
                  VT = np.zeros((n*m,1), int)

      return M
```

§6 Hints

1. First hint.
2. Second hint.
3. First hint.

§7 Bibliography

Random text for citing some document in my LaTeX file [**Knu97**].

References

[**Knu97**] Donald E. Knuth. *The Art of Computer Programming*. Addison-Wesley, 1997 (cited p. 9)